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Reproductive Biology of the Smoothtail Spiny Lobster (*Panulirus laeviscauda*) to Support Fisheries Management in Southeastern Brazil

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ABSTRACT

The smoothtail spiny lobster (*Panulirus laeviscauda*) is one of the main target species for fisheries in Brazil. We aimed to estimate reproductive parameters of this species in support of a minimum legal size limit for the region. A sample of 508 individuals ranged 13.75–100 mm in cephalothorax length (CL) from the southern coast of Espírito Santo, Brazil, during 2020–2021. Males were more abundant than females. Females reached morphological maturity before males, at 68.4 mm CL, for both sexes. Reproduction was between November and March, and some large females were capable of spawning more than once during the reproductive period. Because large adult females could spawn more than once per season, we recommend increasing the legal minimum capture size of *P. laeviscauda* to 70 mm CL in Brazil.

1 | Introduction

Lobsters have been exploited for over a century worldwide as marine fishery resources and are one of the most profitable fisheries industries, with around 27 targeted lobster species (Cruz et al. 2021; Food and Agriculture Organization of the United Nations [FAO] 2022; Pereira and Josupeit 2017). In 2013, global lobster production of all species was 231,968 t, of which two families, Nephropidae and Palinuridae, accounted for > 80% of the lobster biomass landed in 2013 (Ward and Phillips 2013). Brazil is noteworthy in world lobster production (Alencar et al. 2021; Dias-Neto 2008; Oliveira et al. 2015), with an export revenue for spiny lobsters of US\$ 117 million in 2021 (*Panulirus* spp.) that comprised 6000 tons (COMEXSTAT 2022; Nunes & Santander-Neto, submitted).

In Brazil, commercial lobster fishing takes place from the coast of the state of Pará to the coast of the state of Espírito Santo over a 74,607 km² fishing area (Dias-Neto 2008; Fonteles-Filho 2007). Widely distributed throughout shallow waters, lobsters of the *Panulirus* genus are affected by multiple environmental and anthropogenic stress factors that are mainly associated with fisheries, such as high illegal fishing effort, capture of juveniles, non-compliance with fishery regulations, and fisheries overexploitation (Briones-Fourzán and Lozano-Álvarez 2013; Santana et al. 2016).

Spiny lobsters, especially the native lobster *Panulirus meripurpuratus* (sensu *P. argus*, Giraldez and Smyth 2016) and the smoothtail spiny lobster *P. laeviscauda*, have significant fisheries potential. Both are the main targets for artisanal

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fisheries, are the most commercialized species in Brazil, and account for around 80% and 20% of total national lobster production, respectively (Dias-Neto 2008; Santana et al. 2016; Kinas et al. 2020; Gaeta et al. 2022). Native lobsters are more abundant in Northern and Northeastern Brazilian fisheries (Alencar et al. 2021), while smoothtail spiny lobsters are more abundant in the Southeast, mainly in the state of Espírito Santo (Hostim-Silva and Soares 2013). Lobster biomass in Brazil corresponds to 20,000 tons (Aragão 2013; Cruz et al. 2013; Andrade 2015; Aragão and Cintra 2018) with less than 18% of natural stocks suffering from overfishing since the 1980s (Kinas et al. 2020).

The smoothtail spiny lobster (*P. laeviscauda*) occurs in tropical regions from Cuba to Rio de Janeiro, Brazil (Baisre and De Quevedo 1982; Chace Jr. and Dumont 1949; Cruz 2002). It is found from the intertidal zone, sheltered in rock formations, to depths of up to 50 m (Baisre and De Quevedo 1982). Juveniles are found close to the coast and reach ~30 cm in total length when caught in deep areas (Paiva et al. 1971). The species reproduces throughout the year, with multiple matings and spawnings during the same reproductive cycle. In the state of Ceará (Northeast Brazil), reproductive activity is greatest between January and May (Soares and Fonteles-Filho 2000). However, despite its extensive geographic distribution, studies of smoothtail spiny lobster populations, fisheries, and reproductive aspects are restricted to northeastern Brazil (Alves and Tome 1966; Baisre and De Quevedo 1982; Cruz et al. 2013, 2021; Fonteles-Filho 1979, 1986; Ivo and Gesteira 1986; Ivo 2000; Lima and Andrade 2021; Oliveira et al. 2015; Silva et al. 1994; Soares and Fonteles-Filho 2000; Vasconcelos and Vasconcelos 1994).

Our objective was to estimate stock-specific biological parameters related to the reproductive biology of the smoothtail spiny lobster (*P. laeviscauda*) in southern Espírito Santo, including size at maturity, spawning season, and fecundity, to support stock assessments and to establish updated and regionalized management and conservation measures for this important fishery resource in Brazil. Management measures for *P. meripurpuratus* and *P. laeviscauda* have been used in Brazil since 1967 (Cavalcante and Furtado Neto 2012; Dias-Neto 2008, 2017). Current lobster management systems in the country include a closed season between November 1st and April 30th (Brazil, 2021) and a minimum catch size of 65 mm cephalothorax length (CL) for *P. laeviscauda*. However, biological parameters for this species, such as reproductive period and size at sexual maturity, differ among Brazilian regions (Almeida et al. 2021; Soares and Fonteles-Filho 2000). Biological parameters vary spatially in response to local environmental characteristics, fishing pressure, and climate change that causes increasing ocean surface temperatures or food availability changes (Almeida et al. 2021; Atherley et al. 2021; Öndes et al. 2017). Therefore, knowledge of population and reproductive ecology of commercially exploited resources is important because these factors are directly related to stock renewal (Severino-Rodrigues et al. 2012). This information is essential, especially where associations between stock and recruitment are still unknown (Cruz and Bertelsen 2009; Goñi et al. 2003).

2 | Materials and Methods

2.1 | Study Area

The state of Espírito Santo (20°19'S and 40°20'W) is located on the eastern coast of Brazil and encompasses a coastal area of about 411 km (Martins et al. 2009). The 42 fishing communities that are registered in the state are located around 72 fish extraction and landing points along 14 coastal municipalities (Hostim-Silva and Soares 2013; SEAP 2005). The local climate is pseudo-equatorial, with tropical summer rains (October to March) and a dry season in autumn and winter (April to September) (Albino 1999).

2.2 | Data Collection

Smoothtail spiny lobsters were sampled from commercial fleet landings on the southern coast of the state of Espírito Santo, Southeastern Brazil (Figure 1), in the municipalities of Guarapari (20°39'S and 40°30'W), Anchieta (20°43'S; 40°46'W), and Maratáizes (21°2'S, 40°49'W) between April 2020 and August 2021. The commercial fleet uses bottom gillnets at 20–50 m depths. Lobsters were identified based on color, including carapace with red details dorsally; yellow supraorbital horns with black or brown spots; antennules not striped; pereopods with stripes (not continuous) and intense green propodus and dactylus; abdominal somites only white spotted laterally; abdominal somites 1–3 red-brown and 4–6 greenish; pleopods green, white spotted and with yellow border; and membranous part of tail fan in shades of green (darker at extremities) (Giraldes and Smyth 2016). Cephalothorax length (CL, mm) and total weight (TW, g) were measured. Lobster sex was classified based on external secondary characteristics and subsequently confirmed using anatomical gonadal characteristics (Ivo et al. 2013). Macro and microscopic and morphological (Atherley et al. 2021) maturity stages were determined and gonad weight (GW, g) was measured following dissections. Male lobsters were used only for length structure composition assessments, descriptive statistics, and macroscopic maturity size. Ovigerous mass, when present, was removed from females to estimate fecundity. This study was conducted under SISBIO license no. 75244-1 issued by the Brazilian Chico Mendes Institute for Biodiversity Conservation (ICMBio).

The expected 1:1 sex ratio was tested using a chi-squared test ($p < 0.05$), and differences between sexes in CL (mm) and TW (g) were tested using the Mann–Whitney U test. Analysis of covariance was used to compare relationships between CL and TW for each sex ($\alpha = 0.05$).

2.3 | Gonadal Development

The color and appearance (macroscopic characteristics) of each ovary were recorded prior to removal and classified according to color variation, thickness, and texture throughout each developmental stage. The number of females displaying functional reproductive activity characteristics, such as the presence of spermatophores and eggs, was recorded over time. Macro and

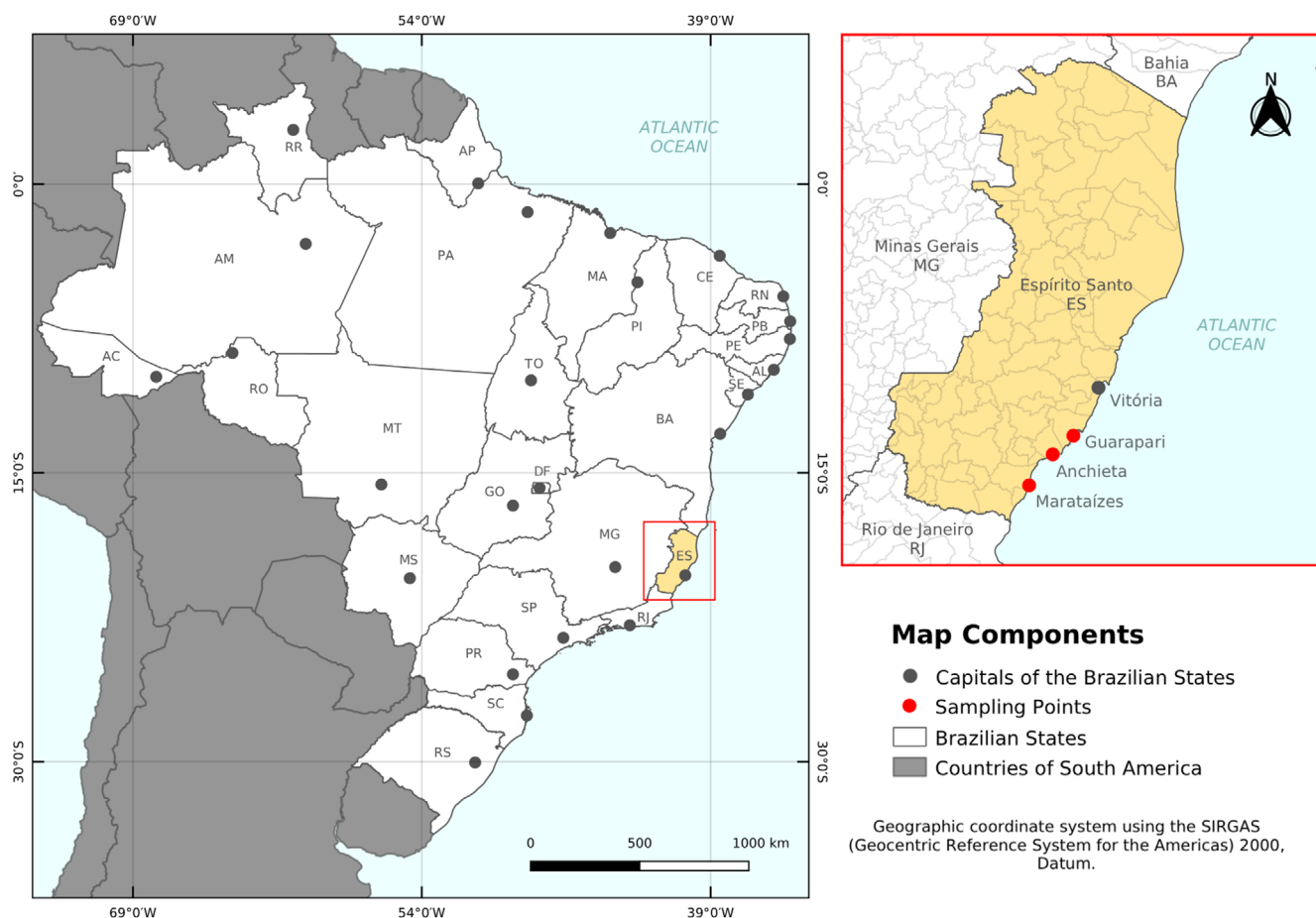


FIGURE 1 | Study area indicating municipalities where smoothtail spiny lobsters (*Panulirus laeviscauda*) were caught off the southern coast of Espirito Santo, Southeastern Brazil, between January 2020 and July 2021.

microscopic evaluations and morphological (secondary sexual characteristics, pleopods—ovigerous setae) female lobster aspects were observed, compared, and used to classify sexual maturity.

Maturity assessments in crustaceans are diverse and categorized with different terminologies (Atherley et al. 2021). Therefore, criteria previously described for lobsters based on macro and microscopic gonadal classification (color, volume, and the presence of blood vessels, among others) were used (Alves and Tome 1966; Goñi et al. 2003; Atherley et al. 2021). Secondary sexual characters were used to determine morphological maturity in crustaceans (Alves and Tome 1966; Goñi et al. 2003; Atherley et al. 2021). Here, the development of endopods on female abdominal pleopods (Kizhakudan et al. 2019) was used to classify morphological maturity as undeveloped, developed, or ovigerous mature (Gregory Jr. and Labisky 1981).

2.4 | Histological Sample Processing

Female gonads were removed intact and sectioned into approximately 3 mm³ samples in the median portion of the ovary. After sectioning, gonads were immediately fixed in Davidson's solution for 24 h, followed by preservation in a 70% alcohol solution until histological processing. Fragments were dehydrated

in an increasing alcohol series (70%, 80%, 90%, and 100%), diaphanized in xylene and embedded in paraffin. Blocks were then cut transversely into 5 µm thick slices using a rotary microtome (LUPETEC MRPSA2016) and stained with hematoxylin and eosin (Junqueira and Junqueira 2012). Maturity stages were determined through germline components observed in photomicrographs obtained with an HD camera and processed through the SIGMA Scan Pro 5.0 software. Development and predominance of germ cells were histologically identified during five gametogenesis phases: I=Immature; II=In early development; III=In final development; IV=Spawning capable; V=Regressing; and VI=Regenerating.

2.5 | Mature Size

The number of mature lobsters (P) in each 5-mm CL class was used to estimate male and female length at maturity, using the logistic equation, $p = 1/[1 + \exp. -(a + bXCC)]$, where a and b are parameters (Goñi et al. 2003; Sparre and Venema 1992). The length at which 50% of females were mature ($L_{50\%}$) was used to define mean length at maturity. Three methods to assess maturity were applied as follows: (1) morphological maturity based on ovigerous setae and endopodia morphology; (2) macroscopic maturity based on macroscopic reproductive system development; and (3) microscopic maturity based on microscopic reproductive system development (Atherley et al. 2021; Goñi

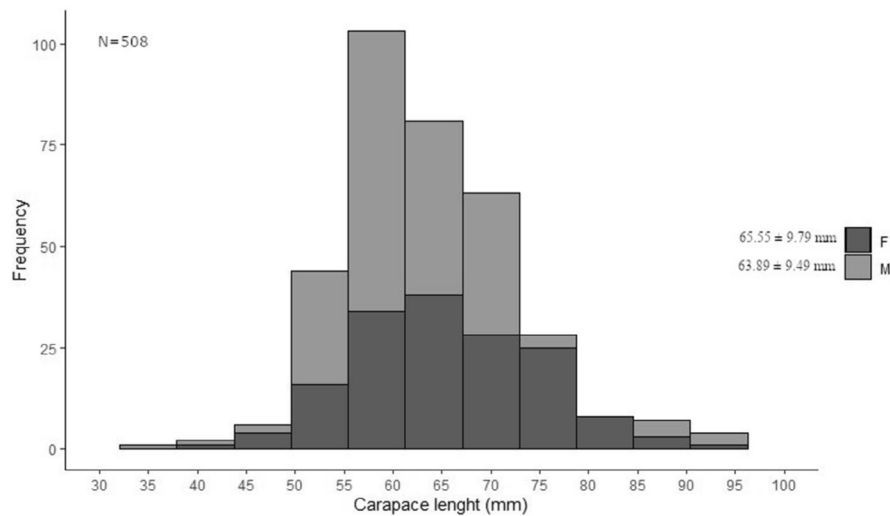


FIGURE 2 | Length frequencies of male and female smoothtail spiny lobsters (*Panulirus laeviscauda*) caught off the southern coast of Espírito Santo, southeastern Brazil, between January 2020 and July 2021.

et al. 2003). For males, only macroscopic maturity was determined through testes analysis. Additionally, $L_{75\%}$ and $L_{99\%}$ (most females of the species reach full maturity) were estimated from the logistic equation.

2.6 | Reproductive Period

Gonad weight (ovary and tests) was used to estimate the gonadosomatic index (GSI) (Minagawa and Sano 1997; Atherley et al. 2021): $GSI = (GW/TW) \times 100$, where TW = total weight (g) and GW = wet gonad weight (g). Mean GSI was estimated to identify potential differences between months of the year using ANOVA (Zar 2010). Mean GSI per month was calculated for each sex and correlated to environmental temperature. Relative maturation stage frequency per month was used as a spawning season indicator. Ocean temperature was used to test for correlations with the reproductive period. Climate data were obtained from climate forecasting platforms (available at <https://pt.weatherspark.com/> and <https://pt.climate-data.org>). Means and standard deviations were estimated, and Pearson's coefficient (r) was used to test for potential correlations.

2.7 | Fecundity

The gravimetric method was used to estimate fecundity by analyzing a ± 0.1 g aliquot of female ovigerous mass. Eggs present in the aliquot were counted, and the total number of eggs in the entire gonad was estimated by expanding the number of eggs in the weight of the aliquot to the weight of the entire gonad. The total egg mass and five aliquots were weighed to the nearest 0.1 g using an analytical scale (Gaeta et al. 2022). Fecundity was estimated following Gaeta et al. (2022), which estimates the total number of eggs (N) of each specimen using the formula $N = (P \cdot n) / p$, where P is the weight of the total egg mass, n is the mean number of eggs in each aliquot, and p is the average weight of each aliquot. The relative fecundity (eggs per body

gram) was estimated (Goñi et al. 2003). Eggs were classified as no pigmentation, initial pigmentation (one or two visible spots), or final pigmentation (several visible spots) to define embryonic development stages (Campbell and Robinson 1983), with the appearance of each stage analyzed over months.

3 | Results

3.1 | Population Structure

Of 508 smoothtail spiny lobster individuals sampled, males were more frequent (344 individuals = 67.71%) than females (164 individuals = 32.28%) ($2.05 \text{ } \delta:1 \text{ } \text{f}; \chi^2 = 287, \text{ df} = 2, p < 0.001$). Female length (range = 35.5–93.3 mm CL; mean $\pm$ SD = 65.55 ± 9.79 mm) varied significantly more than male length (range = 13.75–100.0 mm CL; mean $\pm$ SD = 63.89 ± 9.49 mm; $W = 23,803, p = 0.00193$). Most females ranged 65–70 mm CL and most males ranged 60–65 mm CL (Figure 2). Females weighed significantly more than males (females = 278.0 ± 138 g; males = 226 ± 105 g; Kruskal–Wallis, $H_1 = 15.28, \text{ df} = 1, p < 0.001$). CL was significantly correlated to TW, with males exhibiting negative allometric growth ($PT = 0.00358CC2.65; R^2 = 0.96$) and females exhibiting positive allometric growth ($PT = 0.00019CC3.37; R^2 = 0.88$) (Figure S1). Small males weighed more than females of the same CL, but this trend gradually reversed above CL < 60 mm (Figure S2).

3.2 | Morphology

Endopodia within female abdomens classified as undeveloped were reduced in size, with rounded ends and no filamentous border, similar to a brush (Figure S2A). Females with undeveloped endopodia were smaller (range = 43.45–60.0 mm CL; 53.33 ± 4.5 mm CL) than females with intermediate endopodia (range = 53.7–73 mm CL; 61.89 ± 4.11 mm CL) and had an extended filament on the anterior surface that increased their contact surface (Figure S2B). Mature endopodia in females were

very evident, with a more prominent filament, thereby classifying them as ovigerous or in regression or regeneration stages (Figure S2C; range = 65.50–87.45 mm CL; 76.10 ± 5.30 mm).

Ovaries and testis were similarly paired in a similar “H” shape, with two anterior and posterior lobes in the cephalothorax, between hepatopancreas lobes and joined by a central commissure or “bridge” just above the heart (Figure S3). Oviducts and spermoducts were located in the ventral portion of the cephalothorax, close to the base region of the third female and fifth male pereopod pairs, one connected on each side of lobes to gonopores (genital organ), through which gametes were expelled (Figure S2). Macroscopic classifications of ovarian development stages observed included: immature (Figure S4A), early development (Figure S4B), late development, spawning capable (Figure S4D), regression (Figure S4E,F), and regeneration (Figure S4G,H). Testis development stages observed included immature, developing, and spawning capable (Table S1).

Germ cells observed included: oogonia (OO) of the smallest germ line cells, with a thin layer of basophilic cytoplasm around a well-developed nucleus (Figure S5A); pre-vitellogenic oocytes (PVTG), with nuclear chromatin in a central nucleus surrounded by a thin layer of cytoplasm (Figure S5A); oocytes in early vitellogenesis (VI) containing yolk granules at the cell periphery, an evident nucleus (N), and one or more evident nucleoli (n) (Figure S5A); oocytes in advanced vitellogenesis (AV) containing lipid vacuoles (LV), in addition to yolk granules (YG), with a reduced nucleus (N), and follicular cells (FC) around the oocyte (Figure S4B); mature oocytes (MO) with vesicles implicated in lipid (LV) and protein (YG) vitellogenesis distributed throughout the cytoplasm, with a non-evident and reduced nucleus, and visible zona pellucida (ZR), granulosa (G), and theca (T) (Figure S5C); and oocytes in atresia (AO), reduced in size in atresia (Figure S5D). Germinal ovary zones were apparently active at all development stages, with oogonia and pre-vitellogenic oocytes frequently present at all stages. The main distinguishing characteristic between immature and mature oocyte stages was the presence of more developed oocytes (mature oocyte) in mature individuals. Germ cell organization and distribution in the ovarian stroma distinguished between stages. Microscopic ovarian development stage classifications included: immature (Figure 3A); early development (Figure 3B); late development (Figure 3C); spawning-capable (Figure 3D); regression (Figure 3E); and regeneration (Figure 3F; Table S1).

3.3 | Mature Size

Macroscopic maturity (gonad assessment) was estimated for 119 females (range = 35.55–93.3 mm CL) and 216 males (range = 43.25–100 mm CL). The $L_{50\%}$ for females with ovigerous setae was 59.68 mm CL (CI = 56.61–62.05 mm; $n = 90$; Figure 4C). Length at first macroscopic maturity did not differ significantly between males and females ($L_{50\%} = 68.42$ mm; CI = 67.39–69.36 mm; $n = 327$; ANCOVA, $z = 3.91$, p -value = 0.614; Figure 4A), although males were slightly larger ($L_{50\%} = 68.97$ mm) than females ($L_{50\%} = 67.93$ mm). Microscopic maturity length (histological ovary analysis) was 65.25 mm CL (CI = 63.19–67.41 mm; $n = 74$; Figure 4B). About 96% of lobsters

were similar in maturity status (immature and mature) based on macroscopic and microscopic maturity classifications. Confidence intervals of lengths at maturity overlapped between macro and microscopic estimation methods and differed only for morphological maturity. Macroscopic size at maturity did not differ significantly between males and females ($L_{75\%} = 71.11$ mm CL; $L_{99\%} = 79.67$ mm CL).

3.4 | Reproductive Period

Reproductive period was estimated from 119 females (range = 65.35–93.3 mm CL). Ovigerous females ($n = 14$) were observed from November to February and occasionally in May. Spermatophores were observed in the sternum of all ovigerous females (range = 68.55–93.3 mm CL) from November to February and May. Spermatophores were not intact, but formed a resistant mass in the posterior portion of the sternum, between the third and fifth pair of pereopods.

Macroscopic ovarian development was observed throughout the year (Figure 5). Spawning-capable ovaries were observed throughout the year, except from July to September, with the highest proportion from October to December (25%) and April to June (26%) (Figure 5). Regressing females were frequent from October to June, while regenerating females were observed from January to September, with higher proportions from April to June (21%). The highest proportion of mature ovaries observed (> 53%) was from January to March. From December on, females with spawning-capable ovaries were observed, and the percentage of females in regression gradually increased until February. About 50% of females in regression were ovigerous, with different colored ovigerous masses and embryonic development stages (Figure 6). The ovaries of two ovigerous females (79.4 and 78.0 mm CL) were classified macro- and microscopically as spawning-capable, with pigmented ovigerous masses and advanced embryo development (Figure 6C–E).

Female GSI peaked (1.13) between October and December, gradually decreased after January and was lowest between July and September (Figure 7). Mean GSI of mature females (spawning-capable, regression, and regeneration) was 0.61 ± 0.43 , and the lowest GSI for a regenerating female (0.28; 65.5 mm CL) was similar to that of developing females (Table 1). The highest GSI (4.8; 78 mm CL) was for a female in the regression stage.

During 2020–2021, GSI increased as ocean temperature increased, from $22.8^\circ\text{C} \pm 0.25^\circ\text{C}$, between July and September 2020 (Figure 7), when regenerating females were observed. Between October and May, when spawning is capable, in regression, and ovigerous females were observed, the average maximum ocean temperature reached $25.75^\circ\text{C} \pm 0.35^\circ\text{C}$ and gradually decreased from April on. Females began the regeneration and development process to prepare for the next reproductive season during the decreasing temperature period. The smallest ovigerous female with spermatophores was 68.55 mm CL. Of 14 ovigerous females examined macroscopically and microscopically, 57% ($n = 8$) were spawning-capable between December and February, 36% ($n = 5$) had ovaries in regression between January and May, and

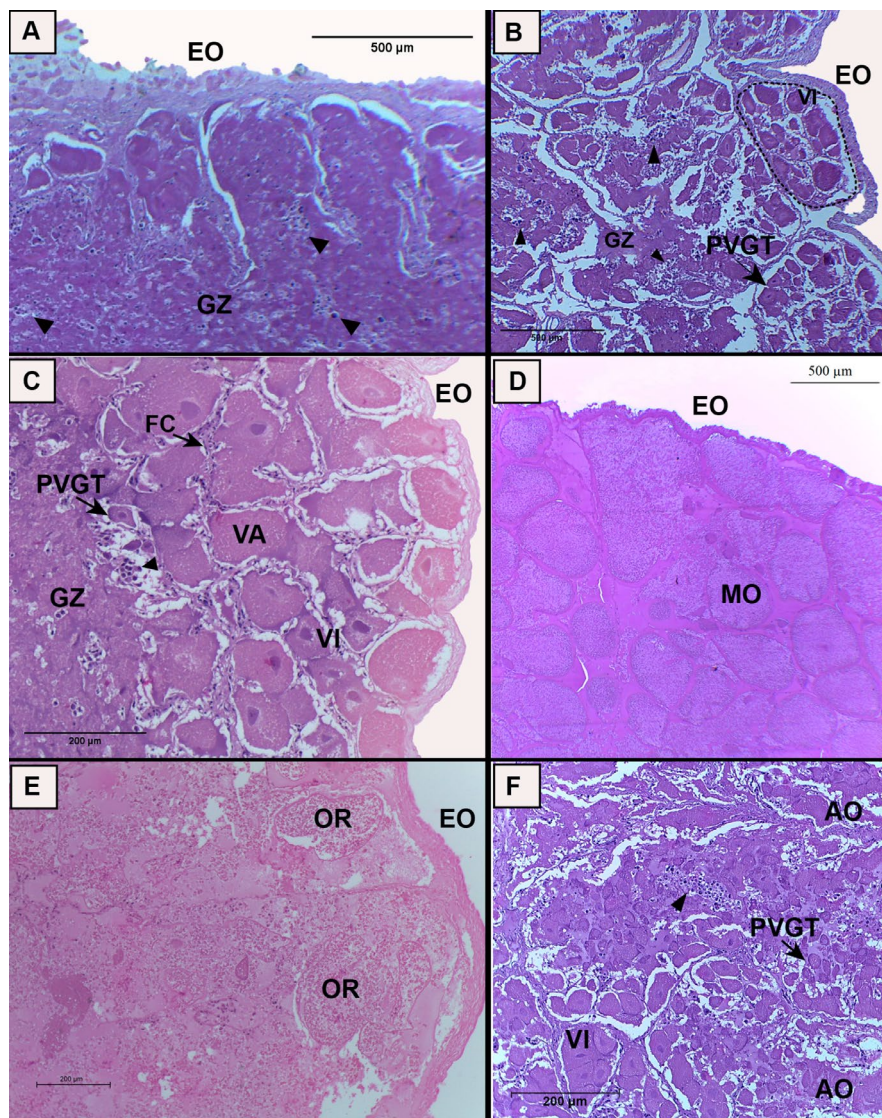


FIGURE 3 | Ovarian development photomicrographs of female smoothtail spiny lobsters *Panulirus laevis* caught in southern Espírito Santo, southeastern Brazil, between January 2020 and July 2021. (A) Immature ovary with the germinal zone (GZ) occupying the entire ovarian parenchyma, consisting predominantly of oogonia (arrowhead). (B) Developing ovary with a reduced germinal zone (GZ), oocyte organization in the ovarian lamellae (dotted line) alongside several cell types: Oogonia (arrowhead), pre-vitellogenic oocytes (PVGT) and early vitellogenesis oocytes (VI). (C) Ovaries in the final development stage present greater germinal zone reduction, consisting of the oocytes observed in the previous stage with the predominance of oocytes in advanced vitellogenesis (AV) surrounded by abundant follicular cells (FC) in the region where AV are concentrated. (D) Spawning-capable ovary, with mature oocytes packed very close to each other. (E) Regressing ovary, containing remaining oocytes (RO), with an emptied appearance. (F) Regenerating ovary containing the same cell types observed in developing ovaries and oocytes in atresia (AO).

7% ($n=1$) were in regeneration in January. Therefore, the reproductive period of *P. laevis* on the southern coast of Espírito Santo began in November, when GSI values gradually increased and spawned ovigerous females were observed, and extended until March, when the last ovigerous females in regression were observed.

3.5 | Fecundity

Ovigerous females were 11% of all females analyzed ($n=119$), with an average weight of 500.91 ± 151.16 g and average carapace length of 78.58 mm. Weight of ovigerous masses ranged 30.31–76.33 g for females of 68.55–86.85 mm CL. Weight of the

ovigerous mass was positively correlated to female CL (ovigerous mass = $3.32 e^{0.032CC}$; $R^2=0.57$; $n=14$; $r=0.75$; $t=3.50$; $df=9$; $p=0.006$) so weight of the ovigerous mass increased proportionally with female size (Figure S6). Female fecundity averaged $209,128 \pm 101,070$ eggs (range = 115,766–487,850 eggs). The smallest female (68.55 mm) had 185,598 eggs, and the largest (93.3 mm) female had 422,836 eggs. The number of eggs was positively correlated to CL (Figure S5; Number of eggs = $20,131 e^{0.031CC}$; $R^2=0.63$; $n=14$; $r=0.80$; $t=4.11$; $df=9$; $p=0.002$). The number of eggs ranged 392–722 per body gram.

Ovigerous females were observed between November and February (Figure 5). The highest numbers of individuals with unpigmented eggs were observed in December (75%), January

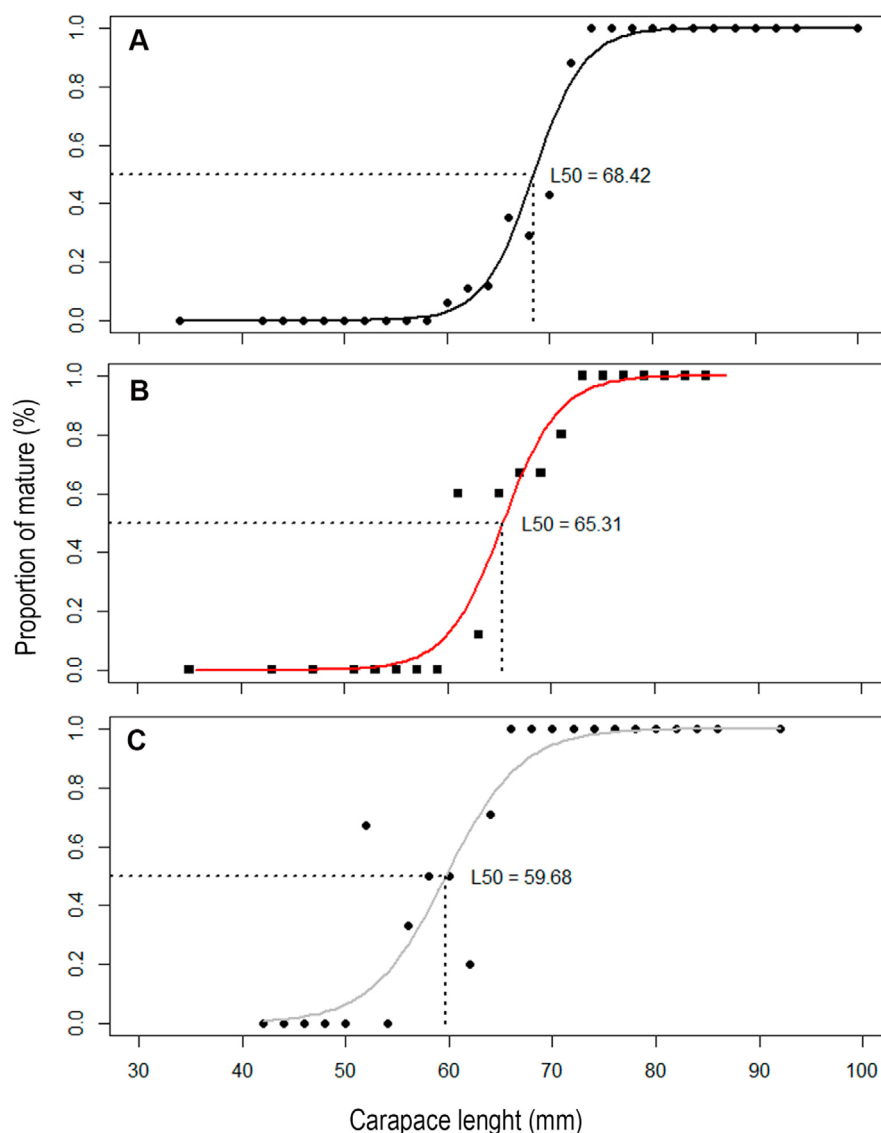


FIGURE 4 | Logistic relationship between carapace length (CL, mm) class and proportion of mature *Panulirus laeviscauda* caught in southern Espírito Santo, southeastern Brazil. (A) Macroscopic maturity for males and females (grouped); (B) microscopic female maturity (histology); and (C) morphological maturity based on female ovigerous setae.

(60%), and February (66.67%) (Figure 7A), with 55% having ovigerous masses with unpigmented eggs, 28% with eggs in the initial pigmentation stage (Figure 7B), and 17% in the final pigmentation stage (Figure 7C).

4 | Discussion

Male and female smoothtail spiny lobsters in our study were caught in different numbers in the region, with a predominance of males, similar to the state of Ceará (Soares and Fonteles-Filho 2000) and the states of Rio Grande do Norte and Bahia/Northern Espírito Santo (Ivo and Gesteira 1995; Alencar et al. 2021). Different sex ratios may be associated with the fact that fisheries in Espírito Santo, for example, operate at greater depths (> 40 m) than in Ceará and Rio Grande do Norte (~25 m) (Ivo 2000). Temperature and depth can influence the sex ratio of lobsters harvested by fisheries, as for male *Homarus americanus* from the northeastern United States,

which prefer warm, shallow waters adjacent to deep, cold areas (Jury et al. 2019). Sex-specific differences in catchability, mortality (Pardo et al. 2015; DFO 2000, 2004), and movement (distribution throughout time and space) of other lobster species may not apply to smoothtail spiny lobsters, which should be studied to identify factors that influence variation in *P. laeviscauda* populations throughout Brazil. Males, although more frequent in our study, were smaller than females potentially due to sexual segregation in different habitats (Hogarth and Barratt 1996; Ebert and Ford 1986). The specimens caught during our study in southern Espírito Santo were smaller than in northern Espírito Santo (males = 85.5 mm; females = 81.2 mm; Ivo and Gesteira 1995). Differences in size between studies may be associated with spatial segregation of life stages because catches in northern Espírito Santo are farther from the coast, where larger lobsters are usually caught (Ivo and Gesteira 1995), whereas catches farther south in our study were in shallower water. However, high fishing effort also leads to a reduction in average capture length, an

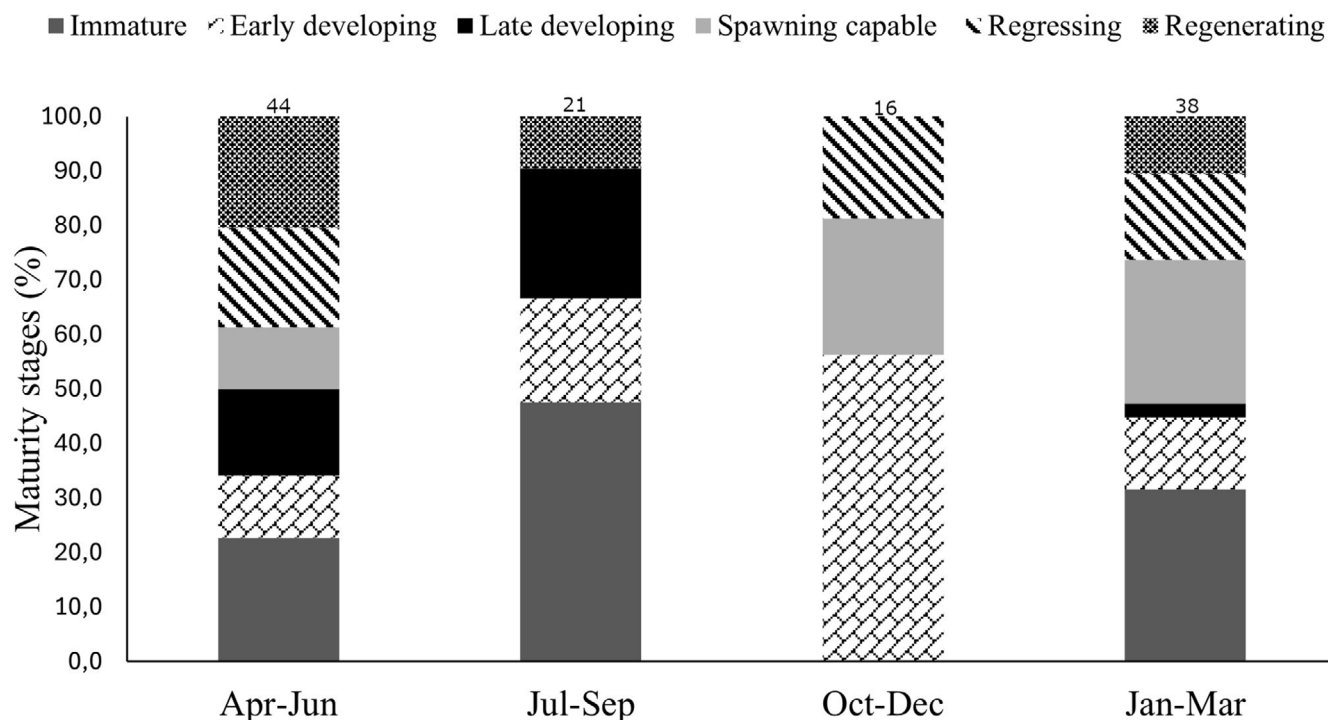


FIGURE 5 | Macroscopic maturation stages of gonads in female smoothtail spiny lobsters *Panulirus laeviscauda* ($n=119$) caught in southern Espirito Santo, southeastern Brazil, between January 2020 and July 2021. The numbers above bars are monthly sample sizes.

indication of overfishing, as in Northeastern Brazil (Ivo and Gesteira 1995; Ivo 2000; Alencar et al. 2021).

In our study in Northeast Brazil, an average lobster was larger than in previous studies in Ceará (males=58.1 mm, females=57.3 mm) and Rio Grande do Norte (males=58.7 mm, females=55.4 mm; Ivo and Gesteira 1995). In this case, size differences are probably not due to distance from the coast because catches were close to the coast. Instead, reductions in *P. laeviscauda* numbers in Northeastern Brazil were associated with high sustained fishing effort in the area that reduced the average catch length (Ivo and Gesteira 1995; Ivo 2000; Alencar et al. 2021). In addition, environmental differences among areas likely caused differences in length structures among populations (Ivo 2000). However, studies of the genetic structure of *P. laeviscauda* populations are still needed to define different populations, if any, and life history of the species in different Brazilian regions (Almeida et al. 2021).

One tool to mitigate overexploitation of lobster stocks is to discriminate mature individuals in management measures. We estimated length at maturity using three different methods to ensure accuracy and precision of estimates for spiny lobsters (Goñi et al. 2003; Chang et al. 2007; Radhakrishnan et al. 2015; Atherley et al. 2021). Characteristics that precede morphological maturity, such as ovigerous setae development and maturation, take place prior to the onset of macro and microscopic maturity processes that are often used to estimate length at first maturity. Similarly, for *Panulirus argus* in the Caribbean, females reached morphological maturity at lengths below the length at maturity because they can apparently molt after developing secondary characteristics but before initiating ovary development (Kulmiye et al. 2006; Atherley et al. 2021). Secondary sexual

characteristics like these are external maturity indicators for some lobster species, such as *Jasus edwardsii* (Hutton, 1875) (Gardner et al. 2005). However, for spiny lobsters, maturity estimates based on ovigerous setae underestimate length at maturity and are not reliable for estimating minimum capture length. Given the similarity in stage classification between established macro and microscopic maturity classes, we used a more conservative estimate to estimate size at maturity of smoothtail spiny lobsters.

The macroscopic length at maturity estimated in our study (68.42 mm) was larger than in northeastern Brazil (48–56 mm CL; Mesquita and Gesteira 1975; and 63 mm CL; Soares and Peret 1998). Differences between studies may be associated with different fisheries practices in coastal waters and large landing variation in Brazil in addition to the influence of biotic and abiotic factors such as recruitment, abundance, temperature, and food availability (De Martini et al. 2003; Cruz et al. 2013). Furthermore, the estimated $L_{50\%}$ and the smallest ovigerous female (68.5 mm CL) we observed were larger than the 65-mm minimum legal length for *P. laeviscauda* in Brazil. Therefore, we recommend a minimum capture size of 70 mm for *P. laeviscauda* in Southeastern Brazil to protect young individuals and ensure adequate recruitment of the species. In the case of a significant local need for recovery of this species, we recommend a more conservative minimum capture size of 72 mm (corresponding to $L_{75\%}$) or 80 mm (corresponding to $L_{99\%}$).

Large female lobsters can spawn more than once during the reproductive period in addition to being more fertile (Cruz and Bertelsen 2009; Dávila et al. 2009). For *P. laeviscauda* in our study, ovigerous females were observed with ovaries containing mature oocytes that were characteristic of the pre-spawning

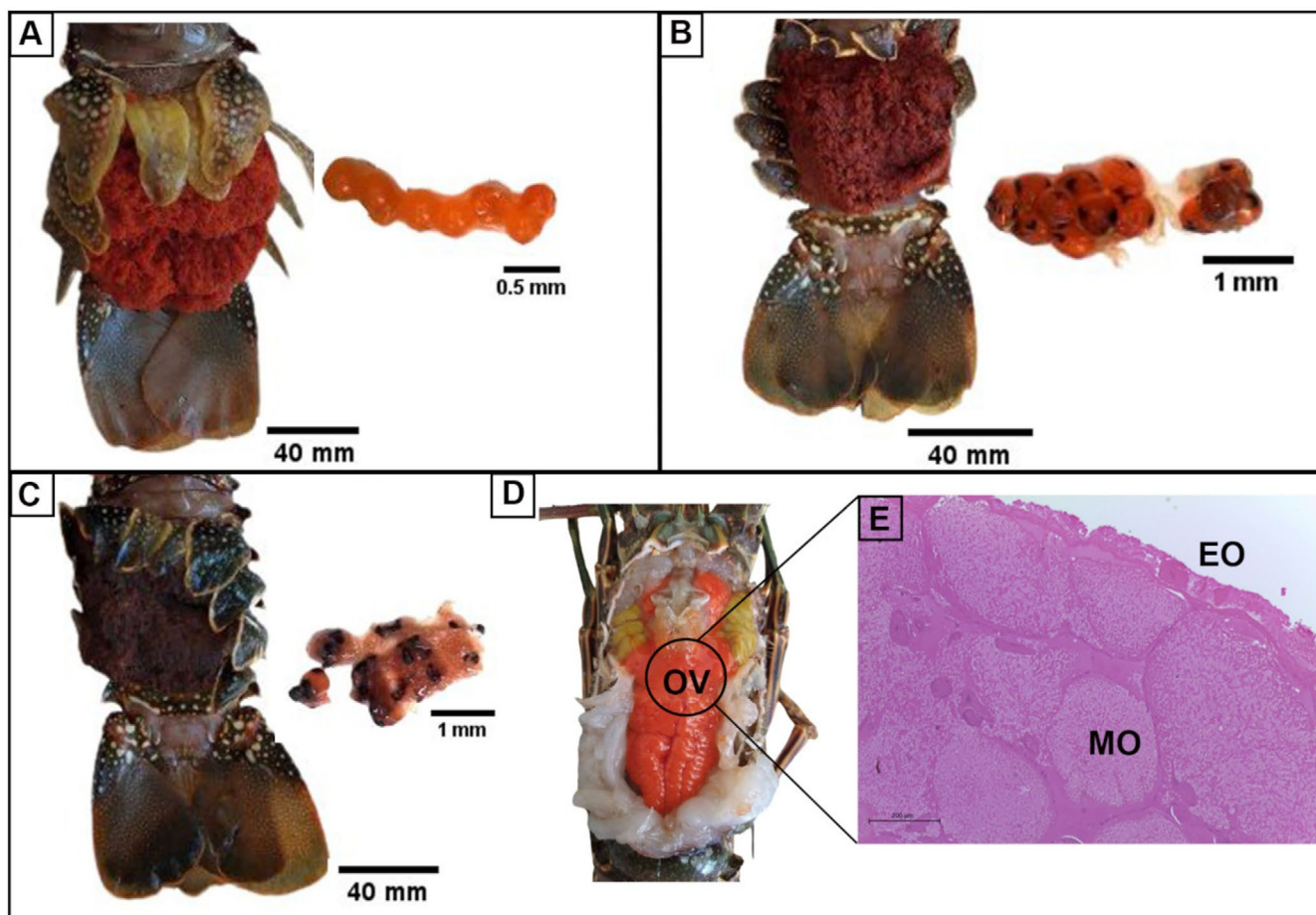


FIGURE 6 | Ovigerous masses in female smoothtail spiny lobsters *Panulirus laeviscauda* during development stages classified according to pigmentation caught in southern Espírito Santo, southeastern Brazil, between January 2020 and July 2021: (A) no pigmentation; (B) initial pigmentation; (C–E) different images of the same individual, where (C) mass presenting its final pigmentation and spawning-capable. (D) Macroscopic views of the ovaries (OV) of an ovigerous female presenting its final pigmentation and spawning-capable. (E) Photomicrograph of a spawning-capable ovary with mature oocytes (MO) packed close to the spawn. OE, ovarian epithelium.

period. Our findings are even more important for overfished populations because the establishment of management measures such as a minimum capture size at lengths greater than $L_{50\%}$ can maintain mature females in the population for more than one spawning period and also favor increased recruitment associated with greater productivity of large females.

Reproductive activity of *P. laeviscauda* in northeastern Brazil (Fonteles-Filho 1979), based on the observation of ovigerous females, occurs most frequently between January and April. In a later study in the same region, the presence of ovigerous females was observed for a prolonged period between January and September (Fonteles-Filho 1979). In the state of Ceará, also in northeastern Brazil, reproductive intensity represented by GSI was highest for the species between February and May (Mesquita and Gesteira 1975). Climatic conditions of the study region differed from those in Northeastern Brazil, which are influenced by the North Brazil and Brazil currents that transport warm waters (average 26°C in winter and 28°C in summer) (Castro Filho and Miranda 1998; Johns et al. 1998; Casey and Cornillon 1999). The state of Espírito Santo is in an upwelling zone that defines a thermal transition zone between tropical waters (north of the state) transported by the Brazil current and cold waters of the South-Central Atlantic originating in the

subtropical convergence region (Santos et al. 2006). Therefore, the *P. laeviscauda* reproductive period occurs throughout the year in northeastern Brazil and peaks only from November to March in Espírito Santo, southeast of the country. Although the reproductive cycle was somewhat similar between studies in northeastern and southeastern Brazil, contributions of small and large females during the reproductive period should be investigated throughout the area because small female *Panulirus* lobsters spawn at the beginning of the reproductive period while larger ones spawn throughout the year (Briones-Fourzán and Lozano-Álvarez 1992).

The reproductive period of smoothtail spiny lobsters we identified in Southeastern Brazil coincided with the closed fishing season for the species in Brazil, where fishing is prohibited from November to April (SAP/MAPA Ordinance No. 688/2022). Therefore, although the closed season was not created considering population parameters of the species in southeastern Brazil, it includes the reproductive period for the species in this region. The reproductive period should also be evaluated for other lobster species that occur in southeastern Brazil (Almeida et al. 2021), such as *P. meripurpuratus*, *P. echinatus*, and *Scyllarides brasiliensis*. Fecundity of *P. laeviscauda* we estimated for the first time for southeastern Brazil was similar to Ceará in

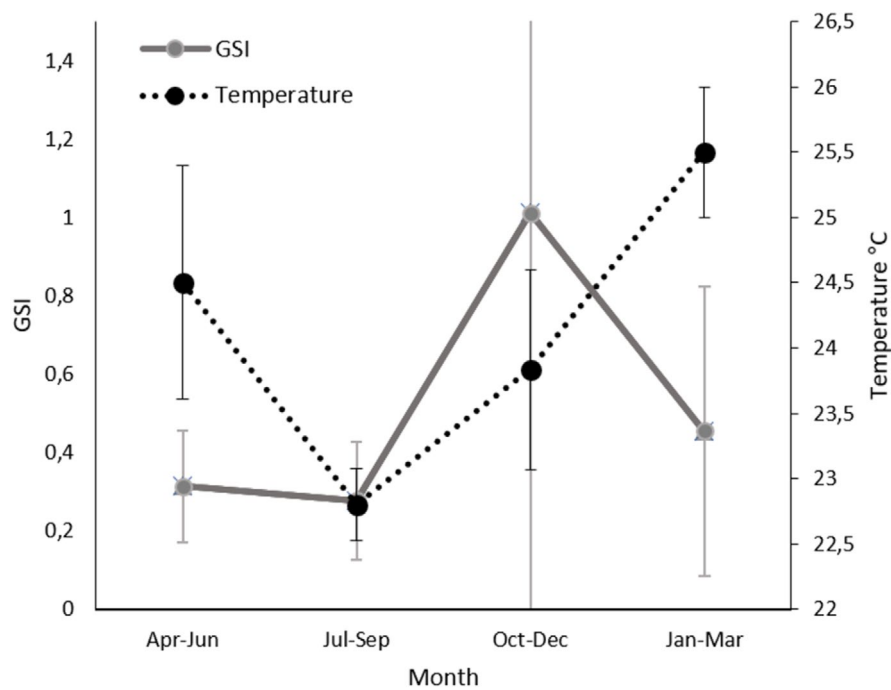


FIGURE 7 | Mean gonadosomatic index (GSI) of female smoothtail spiny lobsters *Panulirus laeviscauda* in southern Espírito Santo (gray) and mean ocean surface temperature off the coast of southern Espírito Santo (black), Brazil, between January 2020 and July 2021.

TABLE 1 | Gonadosomatic index (GSI; $\pm$ and 95% confidence interval) of female smoothtail spiny lobsters *Panulirus laeviscauda* ($n = 112$) caught off the southern coast of Espírito Santo, Southeastern Brazil, from May 2020 to May 2021.

Ovary stage	N	GSI $\pm$ SD	95% CI
Immature	32	0.20 $\pm$ 0.12	0.16–0.24
Developing	32	0.27 $\pm$ 0.09	0.25–0.33
Spawning capable	4	1.11 $\pm$ 0.43	0.42–1.80
Regressing	26	0.97 $\pm$ 1.11	0.52–1.42
Regeneration	18	0.40 $\pm$ 0.16	0.32–0.48

northeastern Brazil, where fecundity declined from 199,104 eggs for females 48.97–92.77 mm CL in 1986 (Ivo and Gesteira 1986) to 175,533 eggs for females 50.00–100.0 mm CL in 1995 (Ivo and Gesteira 1995), before increasing to 200,000–300,000 for females 67–110 mm CL in 2021 (Gaeta et al. 2022). The increase in 2021 was associated with the implementation of a closed season for the species in 2012 (Gaeta et al. 2022), thereby highlighting the importance of such measures and monitoring reproductive parameters over time, especially in different Brazilian regions.

The Smoothtail Spiny lobsters caught in Espírito Santo exhibited an exponential relationship between fecundity and body length as is commonly observed for spiny lobsters (Gaeta et al. 2022). Although fecundity varies with the ability of each female to spawn multiple times, fecundity ranges 300,000–550,000 eggs per female for the Palinuridae family (Pollock 1997), similar to our estimate for *P. laeviscauda*.

Species of high economic and social importance, such as lobsters, require regular updates of their reproductive parameters

to ensure management measures capable of guaranteeing the sustainability of their exploitation. In situations where populations are declining or endangered, it is important to implement management measures, such as those suggested in this study, to establish a minimum capture size or a closed season for the species. These measures aim to alleviate pressure on populations, allow them to reproduce, and ensure the recruitment of the species. Our research highlights the significance of utilizing locally sourced reproductive data for regulatory decisions. Updating new reproductive biology approaches to fishery management protocols must be considered in other countries. Moreover, regions lacking recent data are advised to assess their *P. laeviscauda* population through the methodology presented here.

This first investigation of *P. laeviscauda* in southeastern Brazil enabled us to elucidate reproductive traits of a valuable fishery resource that was paramount for developing management measures aimed at fishery sustainability. The spawning season observed herein coincided with the closed season for the species in Brazil (SAP/MAPA Ordinance No. 221/2021 and No. 688/2022). However, the minimum capture size in Brazil is not sufficient to guarantee capture of only adult lobsters, based on the size at maturity we estimated. We recommend increasing the minimum legal capture size of *P. laeviscauda* to 70 mm CL because large adult females can spawn more than once per season, female fecundity increases with body size, and measures are needed to recover overfished lobster stocks.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.